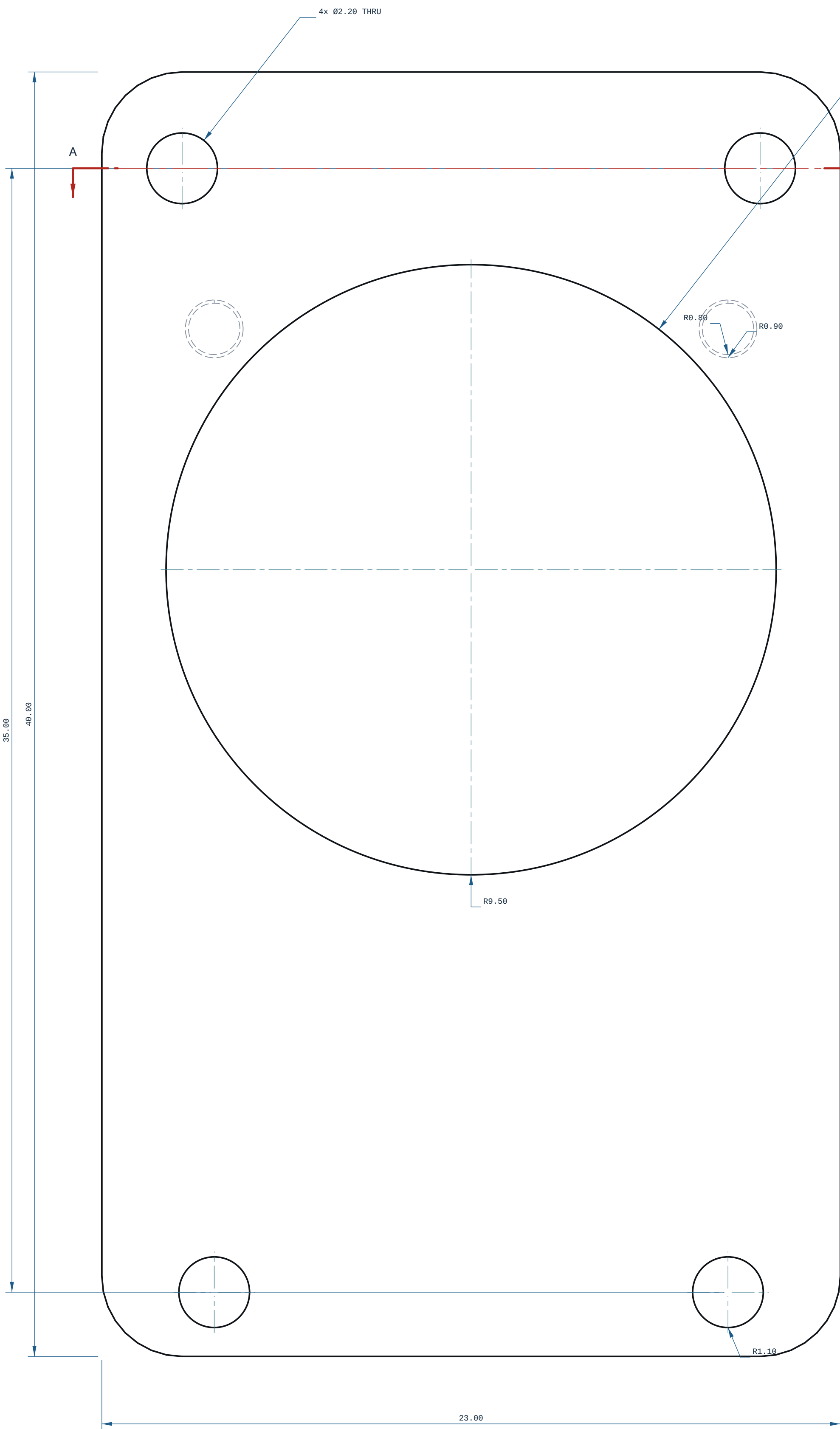


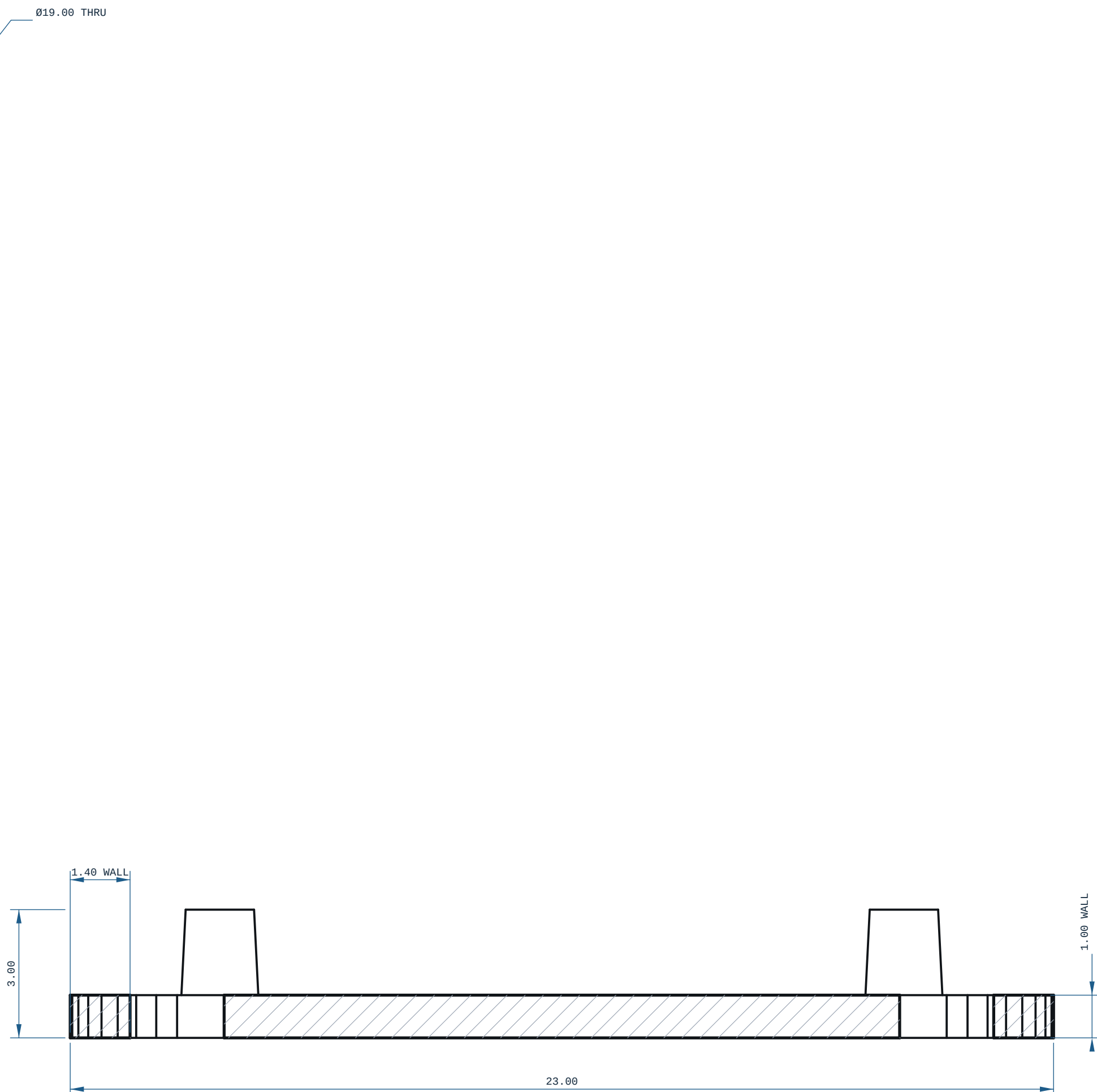
TOP VIEW

X ? Y ?

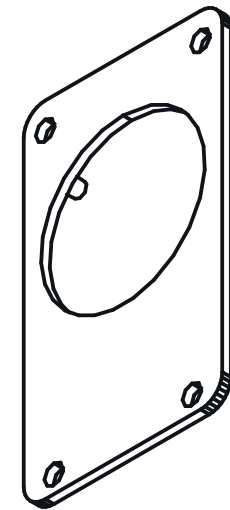


FRONT VIEW

-X ? Z ?



SECTION A-A



ISOMETRIC VIEW

23.00 x 3.00 x 40.00 mm



REFERENCE RENDER (1502) ? rendered off this solid, 900 x 600 px

HOLE TABLE ? 5 POSITIONS

TAG	SIZE	X	Z	C'BORE
A1	Ø2.20 THRU	26.50	12.50	-
A2	Ø2.20 THRU	8.50	12.50	-
A3	Ø19.00 THRU	17.50	0.00	-
A4	Ø2.20 THRU	25.50	-22.50	-
A5	Ø2.20 THRU	8.50	-22.50	-

every row measured off the solid; positions are model coordinates, mm. C'BORE side is the face it opens on; INT = centred in the bore, opening on neither face.

NOTES

- ALL DIMENSIONS IN mm AND DEGREES. EVERY DIMENSION AND EVERY HOLE CALLOUT WAS MEASURED OFF THE SOLID.
- PROCESS: FDM.
- THINNEST WALL, WHOLE SOLID (RAY MINIMUM): 0.9000 mm, AT X=25.518 Y=-14.167 Z=-23.600 ? MEASURED OVER THE WHOLE SOLID BY RAY+EXACT AT 0.4936 mm SAMPLE SPACING (A FEATURE NARROWER THAN THAT CAN BE MISSED); THIS IS A DISTANCE THROUGH MATERIAL, NOT A WALL; AT AN EDGE, A FILLET RUN-OUT OR A CHAMFER TIP IT IS REAL AND UNPRINTABLE BY DEFINITION. DO NOT JUDGE PRINTABILITY BY IT ? THE NEXT NOTE IS THE ONE THAT DOES.
- PRINTABLE MINIMUM WALL: 1.0000 mm, AT X=12.702 Y=-14.500 Z=-17.784 ? MEASURED AS THE SMALLEST MEDIAN WALL OVER A NEIGHBOURHOOD OF 1.4800 mm (>= 8 SAMPLES), RAYS CAPPED AT 4.000 mm, BY cecad.inspect.printable_wall_detail. AGAINST THE 0.8000 mm FLOOR (2 PERIMETERS @ 0.4 mm NOZZLE, cecad/printed.py:801); VERDICT PASS
- DETAIL CALLOUTS (MANUFACTURING-REQUIREMENTS A.7): NONE ON THIS SHEET. DETAIL A dropped: a Ø21.2 mm region needs 20:1 to beat the sheet's 10:1 by 2x and only 5 mm of the bottom band is free ? it would be drawn the same size as the view it points at. THE REGIONS ARE DRAWN AT THE SHEET SCALE IN THE VIEWS ABOVE; NOTHING IS HIDDEN, NOTHING IS ENLARGED. WHAT SETTLES IT: BIGGER PAPER, OR A SECOND SHEET FOR THE DETAILS.
- GENERAL TOLERANCE CANNOT DETERMINE ? SEE PRINT / DFM NOTES UNLESS STATED.
- SECTION A-A AT Z=12.50: THINNEST MATERIAL IN THIS PLANE ONLY 1.00 mm, AT X=28.81, Y=-14.00 ? A SCAN OF ONE CUT, NOT THE PART MINIMUM.
- FRONT VIEW: hidden lines KEPT: without them 30 of 605 edge samples have no ink within 0.35 mm (worst 2.382 mm) and the feature would be missing from the sheet.
- TOP VIEW: hidden lines KEPT: without them 6 of 602 edge samples have no ink within 0.35 mm (worst 0.849 mm) and the feature would be missing from the sheet.

PRINT / DFM

- ORIENTATION: upright, Z-up (23 x 3 x 40 mm)
- THINNEST WALL, WHOLE SOLID (RAY MINIMUM): 0.9000 mm ? a distance through material, NOT a printable wall where it falls on an edge, a fillet run-out or a chamfer tip. Printability is judged on the next line.
- PRINTABLE MINIMUM WALL: 1.0000 mm (smallest median wall over a 1.4800 mm neighbourhood, cecad.inspect.printable_wall_detail) against a 0.8000 mm floor (2 perimeters @ 0.4 mm nozzle): VERDICT PASS
- 5 horizontal bore(s) bridge at the crown: ream any bearing or press-fit bore to size
- FILAMENT: PLA, -1 g (modelled), 200 layers @ 0.2 mm
- TOLERANCE BASIS, IN-HOUSE: CANNOT DETERMINE. Every Bambu on this farm records tolerance_mm: null, cite "no coupon printed and measured on this machine" (ce-machines/bambu-h2s-*, bambu-h2d-*/capabilities.json, read 2026-09-02), and Bambu Lab's own H2D Pro spec sheet carries no dimensional-accuracy line. DIMENSIONS HERE ARE NOMINAL AS MODELLED.
- TOLERANCE BASIS, OUTSOURCED: JLC3DP ±0.3 mm FDM (jlc3dp.com, ce-machines/farm-jlc3dp); Hubs ±0.5%, floor ±0.5 mm (hubs.com/3d-printing/, ce-machines/farm-hubs). Apply the route's figure, not both. The ±0.2 mm in cecad/machining.py fdm is a repo planning constant with no vendor cite and is NOT used here. One printed and measured coupon closes this note.
- NO ISO 286 CLASS IS DECLARED ON ANY BORE: FDM cannot be sized to one (W/P at B3 is a 0.010 mm band). REAM bearing and press-fit bores after printing.

MICRODUCK-BEARING-ROLL

MATERIAL		PROCESS		SCALE	
PLA		FDM		10:1	
CATEGORY		FINISH		SHEET	
plate		AS PRINTED		A1 1/1	
VOLUME (MEASURED)		MASS (ASSUMES PLA)		UNITS	
0.62 cm3		0.8 g		mm / deg	
SIZE (BOX, MEASURED)		DRAWN		PROJECTION	
23.00 x 3.00 x 40.00		ce-cad 2026-09-03		THIRD ANGLE	
SOURCE (DESIGN FILE)					
ce-parts/microduck-bearing-roll/current/cad/part.py					